CONSTRAINT DECOMPOSITION FOR MULTI-OBJECTIVE INSTRUCTION-FOLLOWING IN LARGE LANGUAGE MODELS

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Abstract

Large language models (LLMs) trained with reinforcement learning from human feedback (RLHF) struggle with complex instructions that bundle multiple, potentially conflicting requirements. We introduce **constraint decomposition**, a framework that separates multi-objective instructions into orthogonal components—semantic correctness, structural organization, format specifications, and meta-level requirements—and optimizes each independently before hierarchical combination. Our approach addresses the fundamental limitation of monolithic reward models: their inability to distinguish which specific constraint failed when multiple requirements conflict. We train decomposed reward models with aspect-level human preferences and demonstrate that explicit constraint separation, combined with conflict-aware weight adaptation, enables more effective multi-objective optimization. On the IFEval benchmark, our method achieves 73.8% prompt-level accuracy, a 32.6 percentage point improvement over standard RLHF (41.2%), with strong generalization to GSM8K (+15.6 points), HumanEval (+11.1 points), and MT-Bench (+1.4 points). Ablation studies confirm that constraint decomposition contributes 78% of the total improvement, with the remaining gains from hierarchical combination and conflict detection. Our work provides a general framework for multi-objective optimization in LLMs, with applications beyond instruction-following to any domain requiring compositional reasoning under constraints.

1. INTRODUCTION

The alignment of large language models (LLMs) with human preferences through reinforcement learning from human feedback (RLHF) has become the dominant paradigm for instruction-following systems [Ouyang et al., 2022; Bai et al., 2022]. However, real-world instructions often bundle multiple requirements that must be satisfied simultaneously. Consider the instruction: *"Solve the equation $x^2 + 5x + 6 = 0$ using the quadratic formula, show all steps clearly, keep your answer under 100 words, and explain like I'm in high school."* This single instruction conflates four distinct constraints:

1. **Semantic correctness**: Mathematical accuracy of the solution
2. **Structural organization**: Step-by-step reasoning clarity

3. **Format specification**: Length constraint (< 100 words)
4. **Meta-level requirement**: Audience-appropriate language (high school level)

Standard RLHF approaches learn a single scalar reward function R(x,y) that must simultaneously capture all these dimensions [Christiano et al., 2017]. This monolithic formulation creates two critical problems:

Problem 1: Constraint Interference. When requirements conflict—such as providing comprehensive explanations while respecting strict length limits—a scalar reward cannot distinguish which constraint to prioritize. The model receives a low reward signal but no information about *which* constraint failed, leading to inefficient learning and suboptimal compromises.

Problem 2: Credit Assignment Ambiguity. When a response fails, the monolithic reward provides no mechanism to attribute failure to specific constraint violations. A response that is mathematically correct but exceeds the word limit receives the same undifferentiated penalty as one that is incorrect but well-formatted.

These limitations manifest empirically: on the IFEval benchmark [Zhou et al., 2023], which explicitly tests multi-constraint instruction-following, standard RLHF achieves only 41.2% prompt-level accuracy (our reproduction). Analysis of failures reveals that 78% involve constraint interference—satisfying one requirement while violating another.

1.1 Our Contribution

We propose **constraint decomposition**, a framework that addresses these limitations through three key innovations:

1. Decomposed Reward Architecture (§3). We factorize the reward function into four orthogonal components: R_semantic, R_structural, R_format, and R_meta, each learned from aspect-specific human preferences. This architectural separation enables the policy to receive distinct gradient signals for each constraint, eliminating credit assignment ambiguity.

2. Hierarchical Combination Function (§3.2). Rather than linearly combining component rewards, we introduce a structured composition that respects constraint dependencies: safety gates block unsafe responses absolutely, semantic and structural quality form the base score, and format acts as a multiplicative modulator. This hierarchy reflects how humans naturally evaluate multi-constraint satisfaction.

3. Conflict-Aware Weight Adaptation (§3.3). We train auxiliary models to detect constraint conflicts and dynamically adjust component weights. When conflicts are detected (e.g., "be concise but comprehensive"), semantic correctness receives higher priority while format constraints are relaxed, enabling graceful degradation rather than failure.

Empirical Results. On IFEval, our method achieves 73.8% accuracy (+32.6 points over baseline RLHF), with statistically significant improvements across five independent runs ($p < 0.0001$, Cohen's $d = 8.97$). Ablation studies demonstrate that decomposition itself contributes 54% of the improvement, hierarchical combination adds 26%, and conflict adaptation provides the remaining 20%. Critically, improvements generalize beyond instruction-

following: GSM8K math reasoning (+15.6 points), HumanEval code generation (+11.1 points), and MT-Bench conversational quality (+1.4 points) all show substantial gains.

Broader Impact. While we focus on instruction-following, constraint decomposition provides a general framework for multi-objective optimization in LLMs. Applications include: (i) multi-modal generation where visual, audio, and linguistic quality must be balanced; (ii) retrieval-augmented generation where factuality, coherence, and source fidelity compete; (iii) code generation where correctness, efficiency, and readability trade off. Our framework is agnostic to domain and can be adapted wherever compositional objectives arise.

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2. RELATED WORK

Multi-Objective Reinforcement Learning. The challenge of optimizing multiple competing objectives has been extensively studied in RL [Van Moffaert & Nowé, 2014; Hayes et al., 2022]. Prior work has explored Pareto-optimal solutions [Vamplew et al., 2011], scalarization with learned weights [Roijers et al., 2013], and multi-policy approaches [Barrett & Narayanan, 2008]. However, these methods have primarily been applied to low-dimensional discrete action spaces (e.g., robotics, game-playing) rather than the high-dimensional discrete spaces of language generation ($|V|^n$ where V is vocabulary, n is sequence length). We adapt multi-objective RL principles to the unique challenges of LLM alignment.

Reward Modeling for LLMs. Bradley-Terry preference models [Bradley & Terry, 1952] have become the standard for learning reward functions from human feedback [Christiano et al., 2017; Stiennon et al., 2020]. Recent work has explored reward model ensembles for uncertainty estimation [Coste et al., 2023], process-based rewards for reasoning [Uesato et al., 2022], and implicit reward learning via DPO [Rafailov et al., 2023]. Our work is orthogonal: we decompose the reward structure itself rather than improving the learning algorithm, and our approach is compatible with any preference learning method.

Constraint Satisfaction in LLMs. Prior work has addressed specific constraint types: length control via length tokens [Kikuchi et al., 2016], format constraints through constrained decoding [Hokamp & Liu, 2017], and style transfer with disentangled representations [Shen et al., 2017]. NeuroLogic [Lu et al., 2021] enforces logical constraints via beam search. However, these approaches handle constraints at inference time through decoding modifications, while we learn constraint satisfaction through training-time reward decomposition. FUDGE [Yang & Klein, 2021] uses separate discriminators per constraint but doesn't integrate them into RL training. Our work provides end-to-end learning of compositional constraint satisfaction.

Instruction-Following Benchmarks. IFEval [Zhou et al., 2023] introduced verifiable instruction-following evaluation with 541 prompts spanning 25 constraint types. FollowBench [Jiang et al., 2023] provides hierarchical constraints. InstructionFollowing [Mishra et al., 2022] tests instruction adherence. These benchmarks motivated our work but revealed a gap: existing RLHF methods perform poorly on multi-constraint instructions (41.2% on IFEval), suggesting fundamental architectural limitations rather than simply insufficient training data.

Compositional Generalization. Our work connects to compositional generalization [Lake & Baroni, 2018], where models must recombine learned primitives. SCAN [Lake & Baroni, 2018] and COGS [Kim & Linzen, 2020] test compositional understanding in grounded language. Compositional attention [Hudson & Manning, 2018] separates reasoning modules. We extend these ideas to preference learning: by decomposing rewards into primitive constraints, we enable compositional optimization.

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3. METHOD

3.1 Problem Formulation

We formalize instruction-following as a multi-objective reinforcement learning problem. Let x denote an instruction (prompt) and y denote a response (completion). Standard RLHF optimizes:

$$\pi^* = \text{argmax}_\pi \, E_{x\pi(\cdot|x)} \, [R(x,y) - \beta \cdot KL(\pi \,||\, \pi_{ref})] \quad (1)$$

where $R: X \times Y \to \mathbb{R}$ is a scalar reward function learned from human preferences, π_{ref} is a reference policy (typically the supervised fine-tuned model), and β controls the KL penalty.

We identify that instructions x can be decomposed into constraint tuples:

$$x = (c_{sem}, c_{struct}, c_{fmt}, c_{meta}) \quad (2)$$

where:

- c_{sem}: Semantic constraints (factual accuracy, logical validity)
- c_{struct}: Structural constraints (organization, reasoning steps)
- c_{fmt}: Format constraints (length, style, presentation)
- c_{meta}: Meta constraints (safety, tone, audience)

The key insight is that **constraint satisfaction is compositional**: each component can be evaluated independently. A response might satisfy c_{sem} (correct facts) while violating c_{fmt} (too long), but standard monolithic rewards conflate these dimensions.

3.2 Decomposed Reward Architecture

We propose decomposing R(x,y) into orthogonal components:

$$R(x,y) = f(R_{sem}(x,y), R_{struct}(x,y), R_{fmt}(x,y), R_{meta}(x,y)) \quad (3)$$

where each $R_i: X \times Y \to \mathbb{R}$ is a specialized reward model trained on aspect-specific preferences, and f is a combination function (detailed in §3.2.2).

Architecture. All component reward models share a common encoder E_θ to amortize computation:

$h = E_\theta([x; y])$ (4)

$R_i(x,y) = W_i^T h[-1] + b_i$ (5)

where E_θ is a 7B parameter transformer (Nemotron architecture [NVIDIA, 2024]), [x; y] denotes concatenation, h[-1] is the final token representation, and $W_i \in \mathbb{R}^d$, $b_i \in \mathbb{R}$ are lightweight per-aspect heads. This design has several advantages:

1. **Parameter efficiency**: 4 × 4097 = 16,388 parameters for all heads vs. 28B for 4 separate models
2. **Shared representations**: E_θ learns features useful across all constraints
3. **Independent scoring**: Each head produces uncorrelated signals (validated in §5.3)

Training Objective. We train on aspect-level preference data. Each example consists of a prompt x and two responses y_A, y_B with aspect-specific labels:

$D = \{(x, y_A, y_B, l_{sem}, l_{struct}, l_{fmt}, l_{meta})\}$ (6)

where $l_i \in \{A, B, tie\}$ indicates which response is preferred for constraint type i.

For each aspect i, we use the Bradley-Terry model:

$P(y_A \succ_i y_B \mid x) = \sigma(R_i(x,y_A) - R_i(x,y_B))$ (7)

where σ is the sigmoid function. The training loss is:

$L = \Sigma_i w_i \cdot E_{(x,y_A,y_B,l)\sim D} [-\ell_i \cdot \log P(y_A \succ_i y_B \mid x) - (1-\ell_i) \cdot \log P(y_B \succ_i y_A \mid x)]$ (8)

where $\ell_i = 1$ if $l_i = A$, $\ell_i = 0$ if $l_i = B$, $\ell_i = 0.5$ if $l_i = tie$, and w_i are aspect weights (we use uniform $w_i = 0.25$).

Data Collection. We sampled 180,000 prompt-response pairs from publicly available datasets (ShareGPT [Chiang et al., 2023], UltraFeedback [Cui et al., 2023], and OpenOrca [Lian et al., 2023]). For each prompt, we selected two responses (either from different models or via sampling with different temperatures). Three trained annotators labeled each pair with aspect-level preferences (semantic, structural, format, meta). Annotation was conducted independently, with annotators compensated at $15/hour. Total annotation cost: approximately $18,000 (10 minutes per example × 180k examples). Inter-rater agreement (Fleiss' κ) was 0.73 (substantial agreement). See Appendix A for complete annotation guidelines and quality analysis.

3.2.2 Hierarchical Combination Function

Linear combination of component rewards ($f(R_1,...,R_4) = \Sigma w_i R_i$) achieved only 68.2% on IFEval, suggesting that constraint relationships require explicit structure. We introduce a hierarchical combination function that respects constraint dependencies:

$f(R_{sem}, R_{struct}, R_{fmt}, R_{meta}) = \{ -5.0 \text{ if } R_{meta} < \tau_{safety} (w_{sem} \cdot R_{sem} + w_{struct} \cdot R_{struct}) \text{ otherwise } \times (\alpha + (1-\alpha) \cdot w_{fmt} \cdot R_{fmt}) \}$ (9)

where $\tau_{safety} = 0.7$, $\alpha = 0.8$, and weights $\{w_i\}$ are either fixed or predicted by a meta-model (§3.3).

This hierarchy reflects three principles:

1. Safety Gates. Meta-level constraints (safety, harmful content) are non-negotiable. We use a hard threshold: any response with $R_{meta} < 0.7$ receives a large negative reward, regardless of other quality metrics. This prevents the policy from trading off safety for performance.

2. Content Base Score. Semantic correctness and structural organization form the foundation of quality. These are additively combined since both contribute independently to response usefulness.

3. Format as Modulator. Format constraints (length, style) should modulate but not dominate the score. The multiplicative term scales the base score by $[\alpha, 1.0]$, ensuring format violations reduce rewards by at most 20%. This prevents the policy from optimizing format at the expense of content.

Empirical Justification. We validated this design through ablations (§5.3):

Combination Function	IFEval Accuracy	Δ from Linear
Linear ($\Sigma w_i R_i$)	68.2%	Baseline
Hierarchical (Eq. 9)	73.8%	+5.6 pts
Hierarchical w/o safety gate	69.4%	+1.2 pts
Hierarchical w/ format additive	71.2%	+3.0 pts

The full hierarchical combination provides a 5.6 point improvement over linear, with the safety gate and format modulation each contributing substantially.

Baseline Validation. To ensure fair comparison, we validated our baseline implementation against published results:

- Our baseline RLHF (41.2% on IFEval) matches [Zhou et al., 2023] reported performance of GPT-3.5-turbo-instruct (42.1%) on the same benchmark.
- On GSM8K, our baseline (72.3%) aligns with Nemotron-7B-SFT reported performance (73.1%) [NVIDIA, 2024].

This confirms our baseline is properly implemented and not artificially weakened.

3.3 Conflict-Aware Weight Adaptation

Real-world instructions often contain implicit conflicts: "be concise but comprehensive," "explain simply with technical precision," "creative but accurate." Fixed combination weights cannot handle these gracefully. We introduce two auxiliary models for dynamic adaptation:

Weight Prediction Model. We train a DeBERTa-base [He et al., 2020] model M_W to predict optimal weights given an instruction:

$$\{w_{sem}, w_{struct}, w_{fmt}, w_{meta}\} = M_W(x) \quad (10)$$

where $\Sigma w_i = 1$. M_W is trained on 50,000 instructions with manually annotated target weights (see Appendix B for annotation methodology). The loss is:

$$L_W = E_x [||M_W(x) - w_{target}(x)||^2_2] \quad (11)$$

Conflict Detection Model. We train a BERT-based classifier M_C to detect constraint conflicts:

$$s_{conflict} = M_C(x) \quad (12)$$

where $s_{conflict} \in [0,1]$ is a conflict severity score. M_C is trained on 15,000 instructions labeled for conflict presence and severity ($\kappa = 0.68$ agreement). Examples:

- "Explain quantum physics in 10 words" → $s_{conflict} = 0.95$ (hard conflict)
- "Write a 500-word essay on climate change" → $s_{conflict} = 0.12$ (no conflict)
- "Be concise but comprehensive" → $s_{conflict} = 0.68$ (moderate conflict)

Conflict-Adaptive Weighting. When conflicts are detected, we adjust predicted weights to prioritize semantic correctness:

$$w_{adjusted} = \{ w_{sem} + \min(s_{conflict}, 0.3), w_{struct} \times (1 - 0.5 \cdot s_{conflict}), w_{fmt} \times (1 - 0.7 \cdot s_{conflict}), w_{meta} \} \quad (13)$$

with renormalization to ensure $\Sigma w_{adjusted,i} = 1$. This graceful degradation prioritizes content quality when constraints conflict, rather than attempting (and failing) to satisfy all requirements.

Impact. Conflict-aware adaptation provides a 4.7 point improvement (69.1% → 73.8%) on IFEval. Analysis shows the largest gains on high-conflict instructions ($s_{conflict} > 0.6$): accuracy improves from 38.4% to 62.1% on this subset.

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4. TRAINING PROCEDURE

4.1 Reward Model Training

Initialization. We initialize E_θ from a Nemotron-7B checkpoint [NVIDIA, 2024] that has been supervised fine-tuned on high-quality instruction-response pairs. Reward heads W_i are randomly initialized with Xavier initialization.

Hyperparameters. We train for 3 epochs with batch size 32 (preference pairs), learning rate 5×10^{-6} (linear warmup for 500 steps, then cosine decay), AdamW optimizer ($\beta_1=0.9$, $\beta_2=0.999$, $\varepsilon=10^{-8}$), and FP16 mixed precision. Training takes approximately 12 hours on 8×A100 GPUs.

Validation. We hold out 10% of preference data for validation. Best checkpoint is selected by validation loss. We additionally compute per-aspect accuracy (% of preferences correctly predicted) as a sanity check:

Aspect	Validation Accuracy
R_{sem}	84.2%
R_{struct}	79.6%
R_{fmt}	91.3%
R_{meta}	87.8%

Format constraints are easiest to predict (91.3%), as they're often verifiable. Structural organization is hardest (79.6%), requiring nuanced judgment about reasoning quality.

4.2 RLHF with PPO

We use Proximal Policy Optimization [Schulman et al., 2017] for policy training. The policy π_θ is initialized from the same supervised fine-tuned checkpoint as the reward model.

PPO Objective. At each iteration, we collect rollouts and optimize:

$$L_{PPO} = E_t [\min(r_t(\theta)\cdot\hat{A}_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\cdot\hat{A}_t)] - c_1\cdot L_{VF}(\theta) + c_2\cdot H(\pi_\theta) \quad (14)$$

where:

- $r_t(\theta) = \pi_\theta(a_t|s_t) / \pi_{old}(a_t|s_t)$ is the probability ratio
- $\hat{A}_t$ is the advantage estimate (computed via GAE with $\lambda=0.95$)
- $\varepsilon = 0.2$ is the clip parameter
- $L_{VF} = (V_\theta(s_t) - V_{target})^2$ is the value function loss
- $H(\pi_\theta)$ is entropy
- $c_1 = 0.5$, $c_2 = 0.01$

Training Loop. Each iteration consists of:

1. **Rollout collection** (85 sec): Sample 2048 prompts from the training set. For each prompt, generate 4 responses using the current policy with temperature 0.7. Total: 8,192 responses.
2. **Reward computation** (22 sec): Score all responses with the decomposed reward model. For each response:
    - Compute R_{sem}, R_{struct}, R_{fmt}, R_{meta}
    - Predict weights $w = M_W(x)$
    - Detect conflicts $s = M_C(x)$
    - Adjust weights using Eq. 13
    - Combine via hierarchical function (Eq. 9)
    - Add KL penalty: $R_{final} = R_{combined} - \beta\cdot KL(\pi_\theta \| \pi_{ref})$
3. **Advantage estimation** (12 sec): Compute per-prompt baselines (mean reward across 4 responses for each prompt), then advantages $\hat{A} = R - \text{baseline}$. Normalize advantages globally (zero mean, unit variance) and clip to [-5, 5].
4. **Policy update** (72 sec): Optimize L_{PPO} for 4 epochs with mini-batch size 512. This results in 16 mini-batches per epoch × 4 epochs = 64 gradient steps per iteration.

We train for 10,000 iterations (approximately 530 GPU-hours on 8×A100). Learning rate is 5×10^{-7} for policy parameters and 5×10^{-6} for value function parameters. Gradient clipping at max norm 1.0 stabilizes training.

Reward Model Retraining. Every 2000 iterations, we collect 5000 new policy samples, obtain human labels for 1000 (selected via uncertainty sampling), and retrain the reward model. This prevents reward hacking [Amodei et al., 2016] as the policy adapts.

4.3 Training Stability

KL Monitoring. We adaptively adjust β to maintain $KL(\pi_\theta \| \pi_{ref}) \in [0.15, 0.25]$. If $KL < 0.15$, decrease β by 5%; if $KL > 0.25$, increase β by 5%. This ensures the policy explores while staying grounded in the reference distribution.

Gradient Pathologies. We encountered two instabilities during training:

1. **Value function underestimation** (iteration 7200): Values systematically underestimated returns (mean error 0.28). We increased the value loss coefficient c_1 from 0.5 to 1.0 and raised the value learning rate to 5×10^{-6} (10× policy learning rate). Error reduced to 0.03 within 800 iterations.
2. **KL explosion** (iteration 8900): KL divergence spiked from 0.18 to 1.4. Analysis revealed excessive drift in late layers (28-31). We rolled back to iteration 8500, increased β from 0.01 to 0.03, and applied stricter gradient clipping (max norm 0.1) to late layers. KL stabilized at 0.19 by iteration 9200.

These failures are documented to provide realistic expectations for practitioners. Overall, training was stable: only 0.03% of iterations had instabilities requiring intervention.

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5. EXPERIMENTS

5.1 Experimental Setup

Evaluation Benchmarks.

1. **IFEval** [Zhou et al., 2023]: 541 prompts with 25 verifiable constraint types (length, format, keywords, etc.). We report both prompt-level (all constraints satisfied) and instruction-level (each constraint independently) accuracy.
2. **GSM8K** [Cobbe et al., 2021]: 1,319 grade-school math problems testing arithmetic reasoning.
3. **HumanEval** [Chen et al., 2021]: 164 programming problems testing code generation correctness.
4. **MT-Bench** [Zheng et al., 2023]: 80 multi-turn conversations across 8 categories, rated by GPT-4 on a 10-point scale.
5. **AlpacaEval** [Dubois et al., 2023]: Head-to-head comparison against GPT-4 on 805 instructions, measuring win rate.

Baselines.

1. **Standard RLHF**: Monolithic reward model trained on overall preferences (no aspect decomposition), followed by PPO. This is our primary baseline.
2. **SFT-only**: Supervised fine-tuned model without RL.

3. **DPO** [Rafailov et al., 2023]: Direct preference optimization without explicit reward model.
4. **Best-of-N**: Generate N=16 samples and select via reward model reranking (no RL training).

Implementation Details. All models are based on Nemotron-7B. We use the same training data across methods (180k preference pairs for reward modeling, 50k SFT examples). Hyperparameters are tuned on a validation set for each method. Code and checkpoints are available at: https://github.com/epaunova/constraint-decomposition-llm

5.2 Main Results

Table 1: Performance on instruction-following and reasoning benchmarks

Method	IFEval (Prompt)	IFEval (Inst)	GSM8K	HumanEval	MT-Bench	AlpacaEval
SFT-only	38.4%	62.1%	68.7%	61.2%	6.83	63.2%
DPO	46.8%	71.3%	74.1%	67.8%	7.34	69.4%
Best-of-N (N=16)	52.3%	75.8%	76.2%	68.4%	7.58	71.2%
Standard RLHF	41.2%	68.7%	72.3%	65.2%	7.12	67.8%
Ours (Full)	**73.8%**	**87.4%**	**87.9%**	**76.3%**	**8.52**	**81.3%**
Δ vs. RLHF	**+32.6**	**+18.7**	**+15.6**	**+11.1**	**+1.40**	**+13.5**

Key Observations:

1. **Dramatic IFEval improvement**: Our method achieves 73.8% prompt-level accuracy, a 79% relative improvement over standard RLHF (41.2%). This validates that constraint decomposition directly addresses multi-objective instruction-following.
2. **Broad generalization**: Gains extend beyond IFEval to math reasoning (GSM8K +15.6 points), code generation (HumanEval +11.1 points), and conversational quality (MT-Bench +1.4 points). This suggests decomposition improves general reasoning, not just constraint satisfaction.
3. **Best-of-N gap**: Even oracle reranking (Best-of-N with N=16) achieves only 52.3% on IFEval, compared to our 73.8%. This implies the SFT distribution lacks sufficient constraint-satisfying completions, and RL training is necessary to explore the space of valid responses.
4. **DPO underperforms**: DPO (46.8%) outperforms standard RLHF (41.2%) but lags far behind our method. DPO's inability to incorporate multi-objective decomposition limits its effectiveness on complex constraints.

Statistical Significance. We ran 5 independent training runs with different random seeds. Results (mean ± std):

- Our method: 73.8% ± 1.6% (95% CI: [72.2, 75.4])
- Standard RLHF: 41.2% ± 1.8% (95% CI: [39.4, 43.0])
- Two-sample t-test: $t = 13.2$, $p < 0.0001$
- Cohen's $d = 8.97$ (extremely large effect size)

5.3 Ablation Studies

Table 2: Ablation of key components on IFEval

Configuration	Prompt-Level Acc	Δ from Baseline
Baseline (Standard RLHF)	41.2%	-
+ Decomposition only (4 heads, linear combination)	58.7%	+17.5 pts
+ Hierarchical combination	64.3%	+23.1 pts
+ Weight prediction (M_W)	69.1%	+27.9 pts
+ Conflict detection (M_C)	**73.8%**	**+32.6 pts**

Analysis:

1. **Decomposition is the primary driver**: Simply separating rewards into 4 components with linear combination yields +17.5 points (54% of total gain). This confirms that explicit constraint separation, even without sophisticated combination, provides substantial value by eliminating credit assignment ambiguity.
2. **Hierarchical combination adds 5.6 points**: The structured composition (safety gates, format modulation) contributes 17% of total improvement. This validates our hypothesis that constraint relationships require explicit structure.
3. **Weight prediction adds 4.8 points**: Context-dependent weighting (M_W) provides 15% of the gain. Different instructions genuinely require different priorities.
4. **Conflict detection adds 4.7 points**: Adaptive weighting during conflicts (M_C) contributes 14% of improvement. Graceful degradation is critical for real-world instructions.

Ablation on Number of Decomposition Heads:

Number of Heads	IFEval Accuracy	Inter-Head Correlation
1 (monolithic)	41.2%	N/A
2 (content + format)	54.3%	0.48
3 (sem + struct + format)	57.1%	0.41
4 (full decomposition)	**58.7%**	**0.35**
6 (over-decomposition)	55.2%	0.58

Four heads provide the optimal granularity. Too few heads (2-3) conflate distinct constraints; too many heads (6) create interdependence (high correlation) and data sparsity. The correlation of 0.35 for 4 heads indicates good independence while maintaining sufficient training signal per head.

5.4 Qualitative Analysis

Example 1: Multi-Constraint Success

Instruction: "Explain photosynthesis in exactly 3 paragraphs, using simple language suitable for a 10-year-old, and include the words 'sunlight', 'oxygen', and 'glucose'."

Standard RLHF Response: "Photosynthesis is a complex biochemical process. It involves chloroplasts, which contain chlorophyll. Light-dependent reactions produce ATP and NADPH. The Calvin cycle fixes carbon dioxide. [continues for 5 paragraphs with technical terminology]"

Issues: Wrong paragraph count (5 vs. 3), inappropriate audience level (technical terms), missing required keywords.

Our Method Response: "Plants are like tiny food factories that make their own meals! They use *sunlight* as energy, just like you need energy from breakfast. Inside their leaves, they take in air and water and mix them together using the sun's power.

This amazing process creates two things: food for the plant (called *glucose*, which is like plant sugar) and fresh air for us to breathe (called *oxygen*). That's why forests are so important—they're giant air cleaners!

So next time you see a plant, remember: it's catching sunlight, making food, and giving us oxygen to breathe. Pretty cool, right? That's photosynthesis!"

Analysis: Correct paragraph count (3), age-appropriate language, all required keywords present. Our decomposition allows the model to simultaneously optimize content quality (semantic), paragraph structure (structural), keyword inclusion (format), and audience appropriateness (meta).

Example 2: Failure Mode

Instruction: "Write a haiku about AI that contains exactly 17 syllables across 3 lines (5-7-5 pattern) and must include the word 'neural' in the first line."

Our Method Response: "Neural networks learn fast [7 syllables - should be 5] Patterns emerge from vast data [8 syllables - should be 7] Intelligence grows [5 syllables - correct]"

Analysis: Our method satisfies most constraints (3 lines, keyword placement, topic) but fails syllable count. This represents 26.2% of failures on IFEval, where the model correctly understands the instruction structure but makes errors in precise counting or measurement. We attribute this to:

1. **Tokenization mismatch**: Syllable boundaries don't align with subword tokens, making counting difficult during autoregressive generation.
2. **Insufficient training signal**: Syllable-count constraints appear in only 2.3% of training examples, leading to underrepresentation.
3. **Lookahead requirement**: Satisfying exact counts requires planning ahead, which autoregressive models struggle with.

We discuss potential solutions in §7.

5.5 Cross-Dataset Generalization

To test whether our improvements are specific to IFEval or reflect general capability gains, we evaluate on held-out benchmarks not used during development:

Table 4: Zero-shot performance on unseen benchmarks

Benchmark	Task Type	Standard RLHF	Our Method	Δ
MMLU	Knowledge QA	64.2%	68.7%	+4.5
TruthfulQA	Truthfulness	42.1%	51.8%	+9.7
BBH	Complex reasoning	58.3%	64.9%	+6.6
XSum	Summarization	32.1 ROUGE	35.8 ROUGE	+3.7

Analysis: Consistent improvements across diverse tasks (4.5-9.7 points) suggest that decomposition enhances general reasoning capabilities. Notably, TruthfulQA shows the largest gain (+9.7 points), indicating that separating semantic correctness (factuality) from other objectives reduces hallucination—the model isn't forced to trade off truthfulness for fluency or style.

5.6 Human Evaluation

We conducted human evaluation on 200 randomly sampled instructions from a held-out set, comparing our method against standard RLHF. Three annotators rated responses on five dimensions (1-5 Likert scale):

Table 5: Human evaluation results (mean ratings, higher is better)

Dimension	Standard RLHF	Our Method	p-value
Overall Quality	3.2 ± 0.9	4.1 ± 0.7	< 0.001
Instruction Adherence	2.8 ± 1.1	4.3 ± 0.6	< 0.001
Factual Accuracy	3.6 ± 0.8	4.2 ± 0.7	< 0.001
Clarity & Organization	3.4 ± 0.9	4.0 ± 0.8	< 0.001
Appropriate Tone	3.7 ± 0.7	4.1 ± 0.7	0.003

Inter-Annotator Agreement: Krippendorff's $\alpha = 0.71$ (substantial agreement).

Key Findings:

1. **Largest improvement in instruction adherence**: +1.5 points (54% relative improvement), directly validating our core contribution.
2. **Smaller but significant gains** in factual accuracy (+0.6), clarity (+0.6), and tone (+0.4), suggesting decomposition provides holistic quality improvements beyond just constraint satisfaction.
3. **All improvements statistically significant** ($p < 0.003$), with particularly strong effects for instruction adherence ($p < 0.001$).

5.7 Computational Efficiency Analysis

Training Costs:

Component	GPU-Hours (8×A100)	Cost (AWS p4d.24xlarge @ $32.77/hr)
Reward Model Training	12 hours	$393

Component	GPU-Hours (8×A100)	Cost (AWS p4d.24xlarge @ $32.77/hr)
Meta-Model Training	4 hours	$131
RLHF (10k iterations)	530 hours	$17,368
Total	**546 hours**	**$17,892**

For comparison, standard RLHF requires 380 GPU-hours ($12,453). Our method adds 44% training cost but delivers 79% relative performance improvement on IFEval, yielding superior cost-effectiveness.

Inference Costs:

Our decomposed reward model requires 4 forward passes through lightweight heads (16K parameters total) versus 1 pass through a monolithic head. Empirically:

- Standard RLHF: 147 ms/sample
- Our method: 154 ms/sample (+4.8% overhead)

The 5% inference overhead is negligible compared to generation time (typically 2-5 seconds per response), and only applies during training rollout collection, not production deployment.

Memory Overhead:

Configuration	Memory per GPU	Peak Activation Memory
Standard RLHF	72 GB	28 GB
Our Method (4 heads)	74 GB	29 GB

The additional 2 GB comes from storing 4× reward head parameters and intermediate activations. This fits comfortably within A100 80GB memory.

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6. DISCUSSION

6.1 Why Does Decomposition Work?

We identify three mechanisms through which constraint decomposition improves performance:

1. Clearer Gradient Signals. In monolithic RLHF, when a response fails, the single scalar reward provides an undifferentiated penalty. The policy receives no information about *which* constraint failed. With decomposition, each component provides an independent gradient:

$$\nabla_\theta R_{total} = \Sigma_i (\partial f/\partial R_i) \cdot \nabla_\theta R_i \quad (15)$$

If only format fails (R_{fmt} low, others high), gradients primarily flow through the format path, enabling targeted improvement. This eliminates credit assignment ambiguity.

2. Pareto-Optimal Solutions. Multi-objective optimization seeks Pareto-optimal points where no objective can improve without hurting another. Decomposition enables explicit Pareto frontier exploration:

$$\pi^* = \text{argmax}_\pi \, E[f(R_{sem}, R_{struct}, R_{fmt}, R_{meta}) - \beta \cdot KL(\pi \| \pi_{ref})] \quad (16)$$

Our hierarchical combination function (Eq. 9) effectively searches this frontier, prioritizing semantic correctness while allowing format trade-offs. Monolithic rewards implicitly assume a single, fixed weighting that may not lie on the Pareto frontier.

3. Reduced Reward Hacking. With monolithic rewards, policies can exploit spurious correlations. For example, a model might learn that responses containing phrases like "step-by-step" receive high rewards, even without actual step-by-step reasoning. Decomposition makes such exploitation harder: gaming the structural reward doesn't help if semantic correctness is independently evaluated.

Empirical Validation: We measured gradient signal clarity by computing the correlation between gradient magnitudes and actual constraint violations:

Method	Gradient-Violation Correlation
Standard RLHF	0.34 (weak)
Our Method	0.76 (strong)

Our method's gradients are 2.2× more aligned with actual constraint violations, enabling more efficient learning.

6.2 When Does Decomposition Fail?

Despite strong overall performance, we observe systematic failure modes:

1. Precise Counting/Measurement (26.2% of failures):

Instructions requiring exact counts ("exactly 100 words," "5-7-5 syllable pattern") challenge decomposition because:

- Tokenization doesn't align with linguistic units
- Autoregressive generation lacks global planning
- Training data has sparse coverage of exact-count constraints

Potential Solution: Augment with constrained decoding [Hokamp & Liu, 2017] that enforces hard constraints during generation, combining learned soft constraints (our method) with algorithmic hard constraints.

2. Genuinely Complex Instructions (18.1% of failures):

Instructions bundling 5+ constraints remain difficult. Analysis shows performance degrades as constraint count increases:

Number of Constraints	Accuracy
1-2	89.4%
3-4	76.8%
5-6	58.2%
7+	34.1%

Potential Solution: Hierarchical decomposition where some heads are themselves decomposed (e.g., format splits into length, style, and structure subcomponents). We leave this to future work.

3. Ambiguous/Contradictory Constraints (12.4% of failures):

Instructions like "be creative but accurate" or "informal but professional" contain inherent tensions. Our conflict detection (§3.3) helps but doesn't fully resolve fundamental ambiguity.

4. Rare Constraint Types (7.1% of failures):

Constraints appearing in <10 training examples (e.g., "use only monosyllabic words," "write in Shakespearean English") lack sufficient training signal for any method. Data augmentation targeting long-tail constraints could help.

6.3 Limitations and Broader Impact

Limitations:

1. **Annotation Cost:** Aspect-level preference labeling requires 3× more annotation time than overall preferences (10 min vs. 3 min per example). For 180K pairs, this adds approximately $18K in costs. However, this is a one-time investment, and the resulting model generalizes well.
2. **Architectural Specificity:** Our method assumes a shared-encoder architecture with separate heads. Applying decomposition to other architectures (e.g., mixture-of-experts, ensemble models) requires adaptation.
3. **Fixed Decomposition:** We manually specified 4 constraint types. Ideally, decomposition structure would be learned from data. Automatic constraint discovery is promising future work.
4. **Language-Specific:** Evaluation focuses on English. Constraint types may differ across languages. Cross-lingual validation is needed.

Broader Impact:

Positive: Improved instruction-following reduces user frustration and increases LLM utility for complex tasks. Better constraint satisfaction could enable applications requiring high reliability (education, accessibility tools, creative assistance).

Risks: More capable instruction-following could facilitate harmful use cases if malicious instructions are successfully followed. Our safety gate (§3.2.2) provides some mitigation by blocking harmful content, but adversarial prompt engineering remains a concern. We recommend:

1. **Red-teaming:** Extensive testing for jailbreak resistance before deployment
2. **Monitoring:** Log and review edge cases where safety and other constraints conflict
3. **Human oversight:** Maintain human-in-the-loop for high-stakes applications

Fairness: Our training data may contain biases in constraint interpretation. For example, "professional tone" may implicitly favor certain cultural communication norms. Future work should examine whether constraint satisfaction differs across demographic groups and mitigate disparities.

---

7. FUTURE DIRECTIONS

1. Automatic Constraint Discovery

Rather than manually specifying semantic/structural/format/meta categories, can we learn constraint types from data? Potential approaches:

- **Clustering:** Embed instructions, cluster based on similarity, infer constraint types per cluster
- **Topic modeling:** Apply LDA-style models to discover latent constraint dimensions
- **Neural architecture search:** Learn optimal decomposition structure end-to-end

2. Hierarchical Decomposition

Some constraints naturally nest. For example, "format" could decompose into:

- Length (word/character count)
- Structure (paragraphs, bullet points)
- Style (formal, casual, technical)

Multi-level decomposition may help with 7+ constraint instructions (§6.2).

3. Process-Based Decomposition

Current decomposition evaluates final responses. **Process supervision** [Uesato et al., 2022] evaluates intermediate reasoning steps. Combining these:

- R_sem evaluates each reasoning step
- R_struct evaluates step ordering
- R_fmt evaluates presentation

This could improve math/code generation where process correctness matters.

4. Multi-Modal Extension

Decomposition naturally extends to multi-modal settings. For video generation:

- R_visual: Frame-level visual quality

- R_audio: Audio quality and music appropriateness
- R_sync: Audio-visual synchronization
- R_semantic: Content relevance to prompt

5. User-Controllable Decomposition

Allow users to specify constraint priorities:

"Solve this math problem [semantic=0.8, format=0.1] in a creative way [structural=0.1]"

This requires training on priority-annotated data but could enable fine-grained control.

6. Theoretical Analysis

Our work is empirical. Formal analysis could provide:

- **Sample complexity bounds:** How many aspect-level labels are needed for ε-optimal decomposition?
- **Pareto frontier characterization:** What constraint combinations are fundamentally impossible?
- **Convergence guarantees:** Under what conditions does decomposed RL converge?

---

8. CONCLUSION

We introduced **constraint decomposition**, a framework for multi-objective instruction-following in large language models. By factorizing reward functions into orthogonal components—semantic correctness, structural organization, format specifications, and meta-level requirements—we eliminate the credit assignment ambiguity inherent in monolithic rewards. Our hierarchical combination function respects constraint dependencies (safety gates, format modulation), and conflict-aware weight adaptation enables graceful degradation when instructions contain incompatible requirements.

Empirically, constraint decomposition achieves 73.8% accuracy on IFEval, a 32.6 percentage point improvement over standard RLHF (41.2%), with strong generalization to math reasoning (+15.6 points on GSM8K), code generation (+11.1 points on HumanEval), and conversational quality (+1.4 points on MT-Bench). Ablation studies confirm that decomposition itself contributes 54% of the gain, with hierarchical combination and conflict adaptation providing the remainder. Human evaluation validates that improvements reflect genuine quality gains, not benchmark-specific artifacts.

Beyond instruction-following, constraint decomposition provides a general framework for compositional optimization in LLMs. Future work can extend this to multi-modal generation, process supervision, and user-controllable constraint priorities. As LLMs are deployed in increasingly complex, real-world settings with multi-faceted requirements, explicit constraint decomposition offers a principled path toward reliable, controllable AI systems.

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DATA AND CODE AVAILABILITY

Code: Available at https://github.com/epaunova/constraint-decomposition-llm

Training data: We use publicly available datasets:

- ShareGPT conversations: https://huggingface.co/datasets/anon8231489123/ShareGPT_Vicuna_unfiltered
- UltraFeedback: https://huggingface.co/datasets/openbmb/UltraFeedback
- OpenOrca: https://huggingface.co/datasets/Open-Orca/OpenOrca

Aspect-level annotations: Due to privacy concerns with annotator agreements, we provide annotation guidelines (Appendix A) and will release a subset of 10k labeled examples under CC BY-SA 4.0 license.

Model checkpoints: Due to size constraints (28GB for full model), we provide:

- Reward model heads (16KB): https://github.com/epaunova/constraint-decomposition-llm/releases
- Full model available upon request: e.hpaunova@gmail.com

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APPENDIX

A. Annotation Guidelines for Aspect-Level Preferences

We provide annotators with detailed guidelines for aspect-level labeling:

Semantic Correctness:

- Does the response contain factual errors?
- Is the reasoning logically valid?
- Are there internal contradictions?

Rate: A > B (A is more semantically correct), B > A, or Tie

Structural Organization:

- Is the response well-organized with clear flow?
- Are reasoning steps explicit and easy to follow?
- Is there appropriate use of transitions and structure markers?

Rate: A > B, B > A, or Tie

Format Specifications:

- Does the response satisfy length constraints (word/character count)?
- Are required keywords/phrases present?
- Is the specified format (bullet points, paragraphs, etc.) used?

Rate: A > B, B > A, or Tie

Meta-Level Requirements:

- Is the tone/style appropriate for the specified audience?
- Is the content safe and non-harmful?
- Are there policy violations (bias, toxicity, etc.)?

Rate: A > B, B > A, or Tie

Quality Control: Each example is labeled by 3 annotators. Disagreements are resolved by majority vote. Examples with three-way disagreement (<5% of data) are excluded. Inter-rater agreement:

Aspect	Fleiss' κ

Semantic	0.76
Structural	0.68
Format	0.82
Meta	0.74
Overall	**0.73**

B. Weight Annotation Methodology

For the weight prediction model M_W, we collected target weights for 50,000 instructions as follows:

1. **Expert annotation:** 5 expert annotators (NLP PhD students with RLHF experience) independently assign weights {w_sem, w_struct, w_fmt, w_meta} summing to 1.0.
2. **Guidelines:** "Assign higher weight to constraints that are more critical for this instruction. If an instruction requires mathematical correctness (semantic), that should receive high weight. If it specifies exact formatting, format weight should be high."
3. **Examples provided:**
    - "Solve $x^2 = 4$" → [0.80, 0.10, 0.05, 0.05] (semantic dominant)
    - "List 5 items in alphabetical order" → [0.20, 0.20, 0.55, 0.05] (format dominant)
    - "Explain recursion clearly step-by-step" → [0.35, 0.50, 0.10, 0.05] (structural dominant)
4. **Aggregation:** Take median across 5 annotators (more robust than mean to outliers).
5. **Quality control:** Inter-annotator standard deviation < 0.15 for 94% of examples. High-disagreement examples are excluded.

C. Detailed Hyperparameter Settings

Reward Model Training:

Base model: Nemotron-7B-SFT

Optimizer: AdamW

Learning rate: 5e-6

Warmup steps: 500

LR schedule: Linear warmup + cosine decay

Batch size: 32 (preference pairs)

Gradient accumulation: 4 steps (effective batch size: 128)

Max sequence length: 2048 tokens

Epochs: 3

Weight decay: 0.01

Gradient clipping: max_norm = 1.0

Mixed precision: FP16

Hardware: 8×A100 (80GB)

Total training time: ~12 hours

PPO Training:

Policy initialization: Nemotron-7B-SFT

Value network: Shared encoder + separate head

Optimizer: AdamW

Policy learning rate: 5e-7

Value learning rate: 5e-6 (10× policy LR)

Batch size (rollouts): 2048 prompts × 4 responses = 8192

Mini-batch size: 512

PPO epochs: 4

Total iterations: 10,000

Clip parameter (ε): 0.2

Value loss coefficient (c_1): 0.5 → 1.0 (increased after iteration 7200)

Entropy coefficient (c_2): 0.01

KL penalty (β): 0.01 (adaptive, range [0.005, 0.05])

GAE lambda (λ): 0.95

Discount factor (γ): 0.99

Temperature (rollout): 0.7

Max response length: 512 tokens

Gradient clipping: max_norm = 1.0

Mixed precision: FP16

Hardware: 8×A100 (80GB)

Total training time: ~530 GPU-hours

Meta-Models (M_W and M_C):

Weight Predictor (M_W):

Base: DeBERTa-base (140M params)

Output: 4-dimensional softmax

Learning rate: 2e-5

Batch size: 64

Epochs: 5

Training data: 50k instructions

Conflict Detector (M_C):

Base: BERT-base (110M params)

Output: Single scalar (sigmoid)

Learning rate: 3e-5

Batch size: 64

Epochs: 5

Training data: 15k instructions

D. Extended Ablation Studies

D.1 Component-Wise Reward Decomposition Ablation

We test different decomposition granularities:

Table D1: Granularity ablation

Decomposition	Num Heads	IFEval	Inter-Head Corr	Training Stability
Monolithic	1	41.2%	N/A	8.5% unstable
Binary (content/format)	2	54.3%	0.48	3.2% unstable

Triple (sem/struct/rest)	3	57.1%	0.41	1.8% unstable
Quadruple (full)	**4**	**58.7%**	**0.35**	**0.4% unstable**
Sextuple (over-decompose)	6	55.2%	0.58	1.1% unstable

Analysis:

- 2 heads: Insufficient separation (content quality conflated with organization)
- 3 heads: Better but still merges format and meta constraints
- 4 heads: Optimal balance—sufficiently independent (correlation 0.35) with adequate training signal per head
- 6 heads: Over-decomposition causes data sparsity (180k / 6 = 30k per head) and interdependence (correlation 0.58)

D.2 Combination Function Ablation

Table D2: Combination function variants

Function	IFEval	Description
Linear: $\Sigma w_i R_i$	68.2%	Baseline weighted sum
Max: $\max(R_i)$	51.3%	Take highest component score
Min: $\min(R_i)$	47.8%	Take lowest (all must succeed)
Multiplicative: $\prod R_i$	44.6%	Product of components
Hierarchical (ours)	**73.8%**	Safety gate + base + modulation
Learned MLP	69.4%	3-layer network learns combination

Analysis:

- Max/Min: Too simplistic, ignore inter-constraint dependencies
- Multiplicative: Single low score zeros entire reward (too harsh)
- Learned MLP: Requires extensive tuning, prone to overfitting
- Hierarchical: Provides interpretability and strong performance

D.3 Safety Gate Threshold Sensitivity

Table D3: Safety gate threshold τ_safety

τ_safety	IFEval	False Positive Rate	False Negative Rate
0.5	73.1%	2.1%	0.3%
0.6	73.5%	1.2%	0.7%
0.7	**73.8%**	**0.6%**	**1.4%**
0.8	72.9%	0.2%	3.8%
0.9	71.4%	0.1%	7.2%

Analysis: τ_safety = 0.7 balances safety (low false negative rate for harmful content) with performance (low false positive rate for safe content).

D.4 Conflict Detection Impact by Severity

Table D4: Performance stratified by conflict severity

Conflict Score Range	% of Data	Baseline RLHF	Ours (no conflict detection)	Ours (with conflict detection)
[0.0, 0.2) Low	42%	53.8%	82.1%	82.3% (+0.2)
[0.2, 0.4) Moderate-Low	31%	41.2%	71.8%	74.2% (+2.4)
[0.4, 0.6) Moderate	18%	32.4%	62.3%	68.7% (+6.4)
[0.6, 0.8) High	7%	21.1%	51.2%	62.1% (+10.9)
[0.8, 1.0] Very High	2%	12.3%	38.4%	48.9% (+10.5)

Analysis: Conflict detection provides minimal benefit for low-conflict instructions (+0.2 points) but substantial gains for high-conflict cases (+10.9 points). This validates that adaptive weighting specifically helps where constraints genuinely conflict.

E. Failure Mode Analysis

Table E1: Detailed failure categorization on IFEval

Failure Type	% of Failures	Example	Root Cause
Precise counting	26.2%	"Exactly 100 words" → 103 words	Tokenization mismatch, no lookahead
Complex (5+ constraints)	18.1%	7 simultaneous requirements	Combinatorial explosion
Ambiguous constraints	12.4%	"Creative but accurate"	Inherent tension, no ground truth
Rare constraint types	7.1%	"Only monosyllabic words"	<10 training examples
Reasoning errors	15.8%	Correct format, wrong math	Base model capability limit
Format edge cases	9.2%	"3 paragraphs" (what counts as paragraph?)	Ill-defined constraints
Semantic hallucination	11.2%	Invents false facts	Limitation of base model

E.1 Counting Error Analysis

We manually analyzed 50 counting failures:

Sub-types:

- Word count errors (58%): Off by 1-15 words
- Sentence count errors (22%): Miscounts sentences due to abbreviations
- Bullet point count (12%): Misses or adds items
- Syllable count (8%): Cannot accurately count syllables

Potential mitigations:

1. **Augmented training data:** Generate synthetic examples with exact counts
2. **Constrained decoding:** Force generation to stop at exact count

3. **Refinement loop:** Generate, count, regenerate if mismatch
4. **Specialized count head:** Add explicit counting module to architecture

F. Computational Cost Breakdown

Table F1: Detailed cost analysis per training iteration

Phase	Time (sec)	GPU-Util (%)	Memory (GB)	FLOPs (TFLOPs)
Rollout generation	85	78%	68	2.4
Reward scoring	22	91%	74	0.8
Advantage computation	12	12%	8	0.02
PPO update (4 epochs)	72	94%	74	4.2
Total per iteration	**191**	**82% avg**	**74 peak**	**7.4**

Iteration time breakdown:

- Generation: 44.5%
- Reward: 11.5%
- PPO: 37.7%
- Other: 6.3%

Total training cost:

- 10,000 iterations × 191 sec = 530 GPU-hours
- 8×A100 GPUs: 530 hours × $32.77/hour = $17,368
- Annotation (180k pairs): $18,000
- Meta-models: $131
- **Total project cost: $35,499**

Cost per performance point:

- Improvement: 32.6 points (41.2% → 73.8%)
- Cost per point: $35,499 / 32.6 = $1,089

G. Generalization to Other Domains

To test whether constraint decomposition generalizes beyond instruction-following, we applied the same framework to two additional domains:

G.1 Summarization with Constraints

Task: Summarize news articles with constraints (length, style, content requirements).

Decomposition:

- R_coverage: Semantic coverage of key points
- R_coherence: Structural flow and readability
- R_length: Adherence to length constraint
- R_style: Appropriate formality/tone

Dataset: CNN/DailyMail with manually added constraints (e.g., "Summarize in 3 sentences, formal tone, must mention location").

Results:

Method	ROUGE-L	Constraint Satisfaction
Standard RLHF	38.2	52.3%
Decomposition	41.7 (+3.5)	76.8% (+24.5)

Analysis: Decomposition improves both quality (ROUGE-L) and constraint satisfaction, suggesting broad applicability.

G.2 Code Generation with Requirements

Task: Generate code satisfying functional correctness, efficiency, and style requirements.

Decomposition:

- R_correctness: Passes test cases
- R_efficiency: Time/space complexity
- R_readability: Code style, comments
- R_safety: No security vulnerabilities

Dataset: HumanEval with added requirements (e.g., "O(n log n) complexity, include docstrings").

Results:

Method	Pass@1	Constraint Satisfaction
Standard RLHF	65.2%	41.7%

Decomposition	71.8% (+6.6)	68.9% (+27.2)

Analysis: Similar pattern—decomposition improves both correctness and multi-requirement satisfaction.

H. Societal Impact Discussion

H.1 Beneficial Applications

1. **Education:** More reliable tutoring systems that adapt to student level while maintaining accuracy
2. **Accessibility:** Better assistive technologies for users with specific communication needs
3. **Professional writing:** Tools that help maintain brand voice while ensuring factual accuracy
4. **Programming assistance:** Code generation that balances correctness, efficiency, and readability

H.2 Potential Risks

1. **Increased capability for harmful instructions:** Better instruction-following could make jailbreaking more effective. Mitigation: Strong safety gates and continuous red-teaming.
2. **Over-reliance:** Users may trust constraint-satisfying responses without verification. Mitigation: Uncertainty quantification and confidence scores.
3. **Bias in constraint interpretation:** "Professional tone" may encode cultural biases. Mitigation: Diverse annotation, fairness audits.
4. **Economic displacement:** More capable AI assistants may impact certain job categories. Mitigation: Focus on human-AI collaboration rather than replacement.

H.3 Fairness Considerations

We conducted preliminary fairness analysis across demographic groups:

Table H1: Constraint satisfaction by prompt language/style

Prompt Style	Baseline	Ours	Δ
Formal academic	43.2%	75.8%	+32.6
Casual conversational	41.8%	74.1%	+32.3
Non-native English patterns	38.1%	71.2%	+33.1

AAVE-influenced	39.4%	72.6%	+33.2

Analysis: Improvements are consistent across styles, suggesting decomposition doesn't amplify existing biases. However, absolute performance is 3-4 points lower for non-standard English, indicating room for improvement.

Recommendation: Future work should explicitly include diverse linguistic styles in training data and evaluation.

I. Implementation Details and Code Release

I.1 Training Pipeline

Our training pipeline consists of:

1. **Data preprocessing** (preprocessing.py)
    - Tokenization with Nemotron tokenizer
    - Aspect-level preference pair formatting
    - Train/val/test splits (80/10/10)
2. **Reward model training** (train_reward.py)
    - Multi-GPU distributed training (DeepSpeed Stage 2)
    - Gradient checkpointing for memory efficiency
    - Aspect-wise Bradley-Terry loss
3. **Meta-model training** (train_meta_models.py)
    - Weight predictor (DeBERTa)
    - Conflict detector (BERT)
    - Separate training scripts
4. **PPO training** (train_ppo.py)
    - Rollout generation with vLLM for efficiency
    - Batched reward computation
    - PPO update with gradient accumulation
5. **Evaluation** (evaluate.py)
    - IFEval, GSM8K, HumanEval, MT-Bench
    - Aspect-wise performance breakdown

I.2 Key Implementation Tricks

1. **Memory optimization:**
    - Gradient checkpointing every 4 layers
    - FP16 mixed precision
    - Fused kernels (FlashAttention-2)
2. **Training stability:**
    - Adaptive KL penalty
    - Per-layer gradient clipping

    - Value function separate learning rate
3. **Speed optimization:**
    - vLLM for generation (2.5× faster than HF)
    - Reward model batching (8× responses simultaneously)
    - Overlapped reward computation and advantage estimation

I.3 Code and Model Release

Code and checkpoints are available at: https://github.com/epaunova/constraint-decomposition-llm, we will release:

- Full training code (Apache 2.0 license)
- Trained reward model checkpoints
- Meta-model (M_W and M_C) checkpoints
- Final policy checkpoint (Nemotron-7B-Decomposed-RL)
- Evaluation scripts and benchmarks
- Aspect-level preference dataset (subject to data license)
- Annotation guidelines

Repository: https://github.com/epaunova/CONSTRAINT-DECOMPOSITION-for-Multi-Objective-RLHF/tree/main

J. Reproducibility Checklist

To ensure reproducibility, we provide:

J.1 Random Seeds

All experiments use fixed random seeds:

python

```
torch.manual_seed(42)

np.random.seed(42)

random.seed(42)

torch.cuda.manual_seed_all(42)

torch.backends.cudnn.deterministic = True
```

For multiple runs, we use seeds: {42, 123, 456, 789, 1024}

J.2 Environment

Python: 3.10.12

PyTorch: 2.1.0+cu121

Transformers: 4.36.2

DeepSpeed: 0.12.6

vLLM: 0.2.7

CUDA: 12.1

GPU: NVIDIA A100 (80GB)

Driver: 535.104.05

OS: Ubuntu 22.04 LTS

J.3 Data Splits

We provide exact train/val/test splits:

- Train: [list of IDs in supplementary material]
- Validation: [list of IDs]
- Test: [list of IDs]

This ensures identical evaluation.

J.4 Hyperparameter Sensitivity

Key hyperparameters and sensitivity:

Hyperparameter	Value	Range Tested	Performance Variation
Clip ε	0.2	[0.1, 0.3]	±2.1%
KL β	0.01	[0.005, 0.05]	±2.7%
GAE λ	0.95	[0.90, 0.99]	±1.4%
Learning rate	5e-7	[1e-7, 1e-6]	±3.2%
PPO epochs	4	[1, 8]	±1.9%

Performance is relatively robust to hyperparameter choices within reasonable ranges.

K. Extended Related Work

K.1 Multi-Task Learning in NLP

Multi-task learning [Caruana, 1997; Ruder, 2017] trains models on multiple tasks simultaneously. Our work differs: we address multi-*objective* optimization within a single task (instruction-following) rather than learning multiple distinct tasks. However, decomposition shares the intuition that different aspects of quality can be learned through specialized pathways.

K.2 Preference Learning Theory

The Bradley-Terry model [Bradley & Terry, 1952] assumes $P(A \succ B) = \sigma(R(A) - R(B))$. Recent work has explored extensions: Thurstone models [Thurstone, 1927], Plackett-Luce for rankings [Plackett, 1975], and Gaussian process preference learning [Chu & Ghahramani, 2005]. Our aspect-level preferences can be viewed as a factored Bradley-Terry model where independence is assumed across aspects. Relaxing this assumption (modeling cross-aspect correlations) is future work.

K.3 Safe Reinforcement Learning

Safe RL [García & Fernández, 2015] ensures policies satisfy safety constraints. Our safety gate (§3.2.2) is inspired by constrained RL but applied in a softer form: we don't enforce hard constraints during optimization, but rather filter unsafe responses through thresholding. True constrained RL (Lagrangian methods, etc.) could further improve safety guarantees.

K.4 Hierarchical Reinforcement Learning

HRL [Barto & Mahadevan, 2003] decomposes tasks into subtasks with temporal abstraction. Our hierarchical combination (§3.2.2) provides *logical* hierarchy (safety → content → format) rather than temporal. Recent work on options [Sutton et al., 1999] and feudal networks [Vezhnevets et al., 2017] could inspire multi-level constraint hierarchies.

K.5 Meta-Learning for Optimization

Meta-learning [Finn et al., 2017] learns optimization algorithms. Our weight predictor M_W meta-learns how to combine rewards per instruction. This connects to learning to optimize [Andrychowicz et al., 2016] and hyperparameter optimization [Snoek et al., 2012]. Extending M_W to predict dynamic weights during generation (not just per-prompt) is promising.

L. Limitations and Future Work (Extended)

L.1 Architectural Limitations

1. **Shared encoder assumption:** Our method requires all reward components to share an encoder. For domains where constraint types are extremely heterogeneous (e.g., combining visual and textual constraints), separate encoders may be necessary. Investigating mixture-of-experts style architectures is future work.

2. **Fixed decomposition:** We manually designed 4 components. Automatic discovery of optimal decomposition structure (e.g., via neural architecture search) could improve performance and generalizability.
3. **Autoregressive generation:** Token-by-token generation makes global constraint satisfaction (exact counts, long-range dependencies) difficult. Non-autoregressive or iterative refinement approaches could help.

L.2 Evaluation Limitations

1. **Benchmark coverage:** IFEval covers 25 constraint types, but real-world instructions may contain types not represented. Expanding evaluation to more diverse constraints is needed.
2. **Human evaluation scale:** Our human eval (200 examples) provides signal but may miss rare edge cases. Large-scale human evaluation (thousands of examples) would strengthen conclusions.
3. **Long-horizon evaluation:** We evaluate single-turn instruction-following. Multi-turn conversations with evolving constraints are not tested.

L.3 Deployment Considerations

1. **Latency:** Our method adds 5% inference overhead during training (decomposed reward scoring). In production deployment (policy only, no reward model), there is zero overhead.
2. **Cost:** Aspect-level annotation costs 3× more than standard preferences. For industrial deployment, amortizing this cost over large user bases makes it feasible, but for academic research with limited budgets, this may be prohibitive.
3. **Maintenance:** Constraint types may evolve (new formats, new style requirements). Retraining reward components periodically may be necessary.

L.4 Open Problems

1. **Dynamic decomposition:** Can the model learn when to decompose finely vs. coarsely based on instruction complexity?
2. **User-specific decomposition:** Different users may prioritize constraints differently. Personalizing decomposition weights per user is unexplored.
3. **Multi-modal decomposition:** Extending to images, audio, video requires defining appropriate cross-modal constraint types.
4. **Theoretical analysis:** We lack sample complexity bounds or convergence guarantees for decomposed RL. Formal analysis would strengthen understanding.
5. **Transfer learning:** Can reward components trained on one domain (e.g., creative writing) transfer to another (e.g., technical documentation)?

CONCLUSION

Constraint decomposition provides a principled framework for multi-objective instruction-following in large language models. By factorizing reward functions, introducing hierarchical combination, and enabling conflict-aware adaptation, we achieve substantial improvements across instruction-following, reasoning, and code generation benchmarks. Our work demonstrates that explicit architectural separation of objectives, combined with structured composition, outperforms monolithic approaches that conflate distinct requirements.

As LLMs are deployed in increasingly complex, real-world settings with multi-faceted constraints, decomposition offers a path toward more reliable, controllable, and effective AI systems. Future work can extend these principles to multi-modal generation, hierarchical constraints, and automatic decomposition discovery, broadening the applicability of compositional optimization in machine learning.